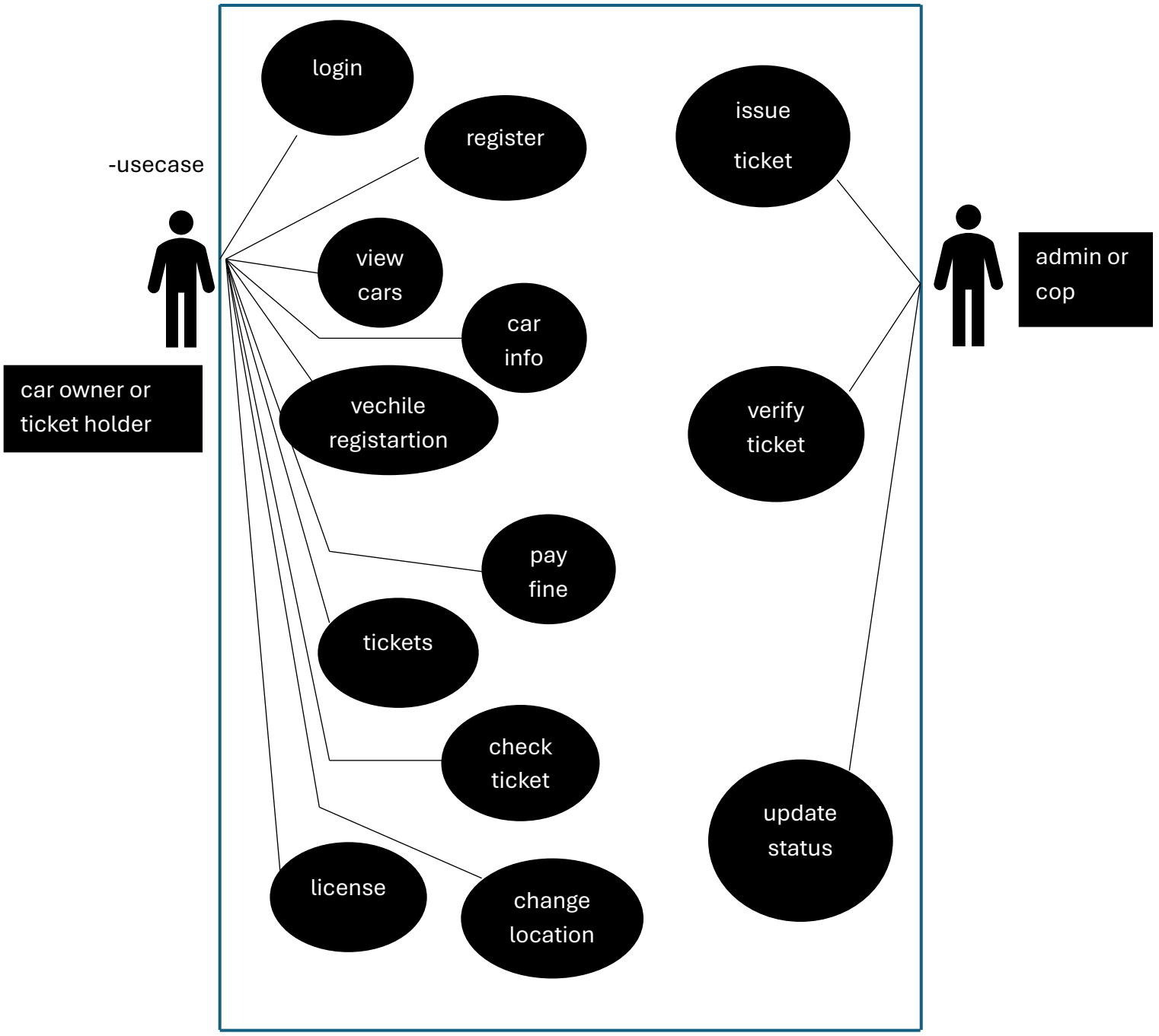
PART 2 :

-usecase



-Actors and goals :

ticket holder	check ticket
car owner	registration
my license	change location
ticket holder	pay the fine
car owner	car info
admin or cop	issue ticket
admin or cop	verify ticket
admin or cop	update ticket status

VEHICLE & TRAFFIC TICKET MANAGEMENT SYSTEM

Use Case Descriptions and Requirements-to-Use-Case Matrix

Project team	Ali, Prajwal, and Sujal
Sections	Use Case Descriptions; Requirements-to-Use-Case Matrix

1. Use Case Descriptions

These descriptions draw on the functions and actor names that appear in the team's documents. Where the source material leaves detailed business rules unspecified, each flow stays at a basic level and should be confirmed with the team.

UC-01: Login

Primary actor: Ticket Holder / Car Owner

Preconditions: The user has a registered account.

Trigger: The user opens the login screen and submits a license number and password.

Main success flow:

• The user enters a license number and password.

• The user selects Login.

• The system validates the credentials.

• The system opens the user's dashboard when the credentials are valid.

Alternative / exception flows:

• If the password is incorrect or the license number is not registered, the system displays a generic login error and does not grant access.

• If required fields are empty, the system displays validation messages.

• After repeated failed attempts, the system may lock or delay further attempts, as proposed in the test cases.

Postconditions: The user is signed in and reaches the dashboard, or remains signed out if authentication fails.

UC-02: Register Account

Primary actor: Ticket Holder / Car Owner

Preconditions: The user does not already have an account.

Trigger: The user opens registration and submits the required personal details, license number, and password.

Main success flow:

• The user enters the requested details.

• The system validates the information and password.

• The system creates the account if the information is valid.

• The system confirms registration.

Alternative / exception flows:

• If the license number is already registered, the system rejects the duplicate account.

• If the password is weak or the confirmation does not match, the system displays a validation message and does not create the account.

Postconditions: A new account exists, or the registration is rejected without creating a duplicate.

UC-03: View Tickets

Primary actor: Ticket Holder

Preconditions: The ticket holder is signed in.

Trigger: The ticket holder opens the Tickets section.

Main success flow:

• The system retrieves tickets associated with the signed-in user.

• The system displays each ticket's number, date, offence, fine amount, and status.

Alternative / exception flows:

• If the user has no tickets, the system displays an empty-state message.

• If the user requests another driver's ticket, the system denies access and does not show the ticket data.

Postconditions: The ticket holder sees their ticket list or an appropriate empty-state message.

UC-04: Check Ticket Details

Primary actor: Ticket Holder

Preconditions: The ticket holder is signed in and has access to the selected ticket.

Trigger: The ticket holder selects a ticket from the ticket list.

Main success flow:

• The user selects a ticket.

• The system checks that the ticket belongs to the signed-in user.

• The system displays the offence, location, date, fine, due date, vehicle plate, and status where available.

Alternative / exception flows:

• If the ticket belongs to another driver, the system denies access.

• If the ticket cannot be found, the system displays an appropriate message.

Postconditions: The ticket details are displayed only to an authorized user.

UC-05: Pay Fine

Primary actor: Ticket Holder

Preconditions: The ticket holder is signed in and has an unpaid ticket.

Trigger: The ticket holder selects Pay for an unpaid ticket.

Main success flow:

• The user selects the unpaid ticket and chooses Pay.

• The user selects an available payment method and authorizes the payment.

• The system processes the payment result.

• When payment succeeds, the system confirms payment, provides a receipt, and updates the ticket status to Paid.

Alternative / exception flows:

• If payment is declined, the system displays an error and leaves the ticket unpaid.

• If the user cancels, the system returns to the ticket without changing its status.

• If the connection is lost during payment, the system checks the actual payment result and avoids a duplicate charge.

• If the ticket is already paid, the Pay action is hidden or disabled.

Postconditions: The ticket is marked Paid after confirmed successful payment; otherwise, its status remains unchanged.

UC-06: View Cars

Primary actor: Car Owner

Preconditions: The car owner is signed in and has vehicle records associated with the account.

Trigger: The car owner opens My Cars.

Main success flow:

• The system retrieves vehicles associated with the owner.

• The system displays the registered vehicles and their plate numbers.

Alternative / exception flows:

• If no vehicles are registered, the system displays an appropriate empty-state message.

Postconditions: The owner can view the list of associated vehicles.

UC-07: View Car Information

Primary actor: Car Owner

Preconditions: The car owner is signed in and can access the selected vehicle.

Trigger: The car owner selects a vehicle from My Cars.

Main success flow:

• The user selects a vehicle.

• The system displays available vehicle information, including make, model, year, plate number, registration status, linked tickets, and accident records.

Alternative / exception flows:

• If the vehicle is not associated with the owner or cannot be found, the system denies access or displays a not-found message.

Postconditions: The vehicle information is displayed to an authorized owner.

UC-08: Register Vehicle

Primary actor: Car Owner

Preconditions: The car owner is signed in and has the required vehicle information.

Trigger: The owner opens vehicle registration and submits vehicle details.

Main success flow:

• The owner enters the plate number, VIN, make, model, and year.

• The system validates the submitted information.

• The system saves the vehicle record.

• The system displays the vehicle in My Cars.

Alternative / exception flows:

• If the plate number is already registered, the system displays an error and does not create a duplicate record.

Postconditions: The vehicle is registered and linked to the owner, or no record is created if validation fails.

UC-09: View License Details

Primary actor: Ticket Holder

Preconditions: The ticket holder is signed in.

Trigger: The user opens the License section.

Main success flow:

• The system retrieves the user's license information.

• The system displays the license number, class, expiry date, and status.

Alternative / exception flows:

• If license information is unavailable, the system displays an appropriate message.

Postconditions: The user can view the available license details.

UC-10: Change Location

Primary actor: Ticket Holder / Car Owner

Preconditions: The user is signed in.

Trigger: The user submits a new address or location.

Main success flow:

• The user enters the new address.

• The system validates the address.

• The system saves the new address and displays confirmation.

Alternative / exception flows:

• If the address is blank or invalid, the system displays a validation error and keeps the previous address.

Postconditions: The new address is saved, or the previous address remains unchanged.

UC-11: Issue Ticket

Primary actor: Admin / Police Officer

Preconditions: The admin or police officer is authenticated and authorized to issue tickets.

Trigger: The officer opens the issue-ticket function.

Main success flow:

• The officer enters the vehicle plate, offence, location, and fine amount.

• The system validates the ticket information and vehicle record.

• The system creates the ticket.

• The ticket becomes visible to the relevant vehicle owner or ticket holder.

Alternative / exception flows:

• If the plate number is unknown or not registered, the system displays an error and does not create a ticket.

• If a normal driver attempts to issue a ticket, the system denies access.

Postconditions: A valid ticket is stored and associated with the relevant vehicle or ticket holder.

UC-12: Verify Ticket

Primary actor: Admin / Police Officer

Preconditions: The admin or police officer is authenticated and authorized.

Trigger: The officer enters a ticket number into the verification function.

Main success flow:

• The officer enters the ticket number.

• The system searches for the ticket.

• The system displays the ticket details and current status when found.

Alternative / exception flows:

• If the ticket number does not exist, the system displays a not-found message.

Postconditions: The officer can see the result of the ticket verification.

UC-13: Update Ticket Status

Primary actor: Admin / Police Officer

Preconditions: The admin or police officer is authenticated, authorized, and the ticket exists.

Trigger: The officer selects a ticket and submits a status update.

Main success flow:

• The officer opens the ticket record.

• The officer selects a status such as Paid, Cancelled, or Disputed, as described in the test cases.

• The system validates and saves the update.

• The updated status is shown to the ticket holder.

Alternative / exception flows:

• If a normal driver attempts to update the status, the system refuses the action and leaves the status unchanged.

• If the ticket cannot be found or the update is invalid, the system does not save the change.

Postconditions: The ticket status is updated for an authorized action, or remains unchanged if the action is rejected.

2. Requirements-to-Use-Case Matrix

Requirement ID	Functional requirement	Use case(s)
FR-01	The system shall authenticate a registered user using a license number and password.	UC-01
FR-02	The system shall allow a new user to register an account using valid personal and license information.	UC-02
FR-03	The system shall allow a ticket holder to view tickets associated with their account.	UC-03
FR-04	The system shall allow an authorized ticket holder to view details of a selected ticket.	UC-04
FR-05	The system shall allow a ticket holder to pay an unpaid fine and receive payment confirmation.	UC-05
FR-06	The system shall allow a car owner to view vehicles associated with their account.	UC-06
FR-07	The system shall display available details for a selected vehicle.	UC-07
FR-08	The system shall allow a car owner to register a vehicle with valid vehicle details.	UC-08
FR-09	The system shall display the user's license number, class, expiry date, and status when available.	UC-09
FR-10	The system shall allow a signed-in user to update their address or location with valid information.	UC-10
FR-11	The system shall allow an authorized admin or police officer to issue a ticket for a valid vehicle record.	UC-11
FR-12	The system shall allow an authorized admin or police officer to verify a ticket using its ticket number.	UC-12
FR-13	The system shall allow an authorized admin or police officer to update a ticket's status.	UC-13
FR-14	The system shall prevent users from viewing ticket records belonging to other drivers without authorization.	UC-03, UC-04
FR-15	The system shall prevent duplicate account, vehicle, or plate registrations.	UC-02, UC-08
FR-16	The system shall reject unauthorized attempts to issue tickets or change ticket status.	UC-11, UC-13
FR-17	The system shall handle failed, cancelled, and interrupted fine payments without incorrectly marking a ticket as paid.	UC-05

Traceability Notes

• The current team diagram lists functions including login, registration, checking tickets, viewing car information, vehicle registration, paying fines, changing location, issuing tickets, verifying tickets, and updating ticket status.

• View Tickets, View Cars, and My License are additionally covered by Prajwal's test cases; they appear here as separate use cases so the descriptions can map to the existing tests.

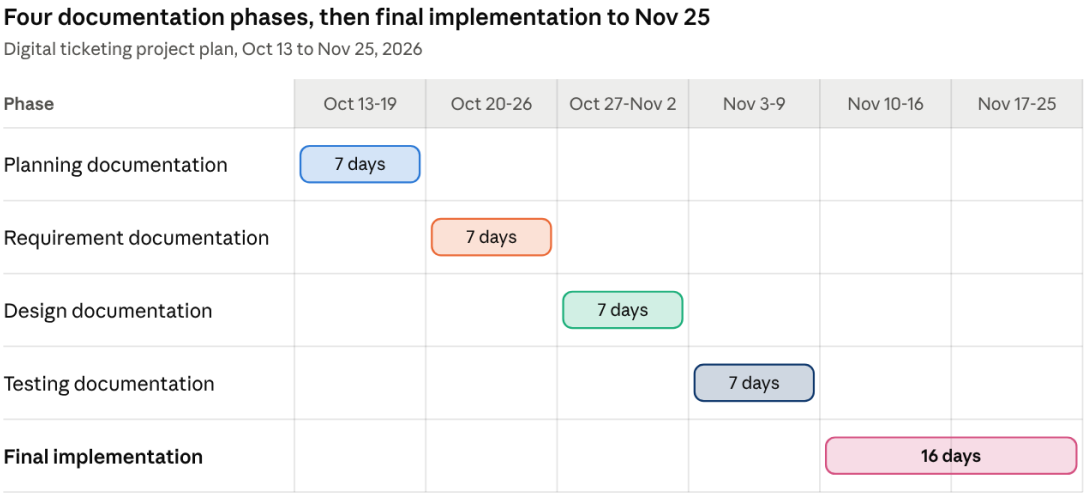
• Where the provided documents do not clearly distinguish which account role performs a shared function, the actor label 'Ticket Holder / Car Owner' is used. Actor permissions should be confirmed with the team.

• If the professor provides numbered requirements, replace the proposed FR-01 to FR-17 IDs with the official requirement IDs and update the matrix accordingly.

Digital Ticketing System: Gantt Chart

The Gantt chart follows the six-week plan from Oct 13 to Nov 25, and the test cases cover every use case in Part 2.

Gantt chart



Gantt chart · 5 phases, Oct 13 to Nov 25

The four documentation phases take one week each, then final implementation takes the last two weeks. The bars in your PDF did not come through as text, so I assumed one phase per weekly column; tell me if yours differ.

Test cases

33 test cases, grouped by use case, cover login, tickets, payment, vehicles, licence, location and the admin or cop functions.

ID	Use case	Test scenario	Steps / input	Expected result	Priority
TC-01	Login	Valid credentials	Enter a valid license number and password, tap Login	Driver reaches the dashboard	High
TC-02	Login	Wrong password	Enter a valid license number and a wrong password	Error message shown, no access granted	High
TC-03	Login	Unregistered license	Enter a license number with no	Generic login error, no account details	High

ID	Use case	Test scenario	Steps / input	Expected result	Priority
		number	account	revealed	
TC-04	Login	Empty fields	Tap Login with both fields blank	Required-field messages shown, nothing submitted	Medium
TC-05	Login	Repeated failed attempts	Enter a wrong password 5 times in a row	Account is locked or delayed and the driver is told why	Medium
TC-06	Register	New account, valid data	Enter license number, personal details and a strong password, submit	Account created, driver can log in	High
TC-07	Register	License number already registered	Register again with an existing license number	Error says the account exists, no duplicate created	High
TC-08	Register	Weak or mismatched password	Enter a short password, or two different confirmation values	Validation error, account not created	Medium
TC-09	View tickets	Driver with several tickets	Log in, open Tickets	List shows ticket number, date, offence, amount and status for each	High
TC-10	View tickets	Driver with no tickets	Log in with an account that has none	Empty-state message shown, no error	Medium
TC-11	Check ticket	Open ticket details	Tap a ticket in the list	Offence, location, date, fine, due date and plate are shown	High
TC-12	Check ticket	Another driver's ticket	Request a ticket ID that belongs to a different license	Access denied, no data shown	High
TC-13	Pay fine	Pay with Apple Pay	Select an unpaid ticket, tap Pay, choose Apple Pay, authorize	Payment confirmed, receipt shown, status changes to Paid	High

ID	Use case	Test scenario	Steps / input	Expected result	Priority
TC-14	Pay fine	Pay with Google Pay	Select an unpaid ticket, tap Pay, choose Google Pay, authorize	Payment confirmed, receipt shown, status changes to Paid	High
TC-15	Pay fine	Payment declined	Pay with a card or wallet that is declined	Error shown, ticket stays Unpaid, no charge made	High
TC-16	Pay fine	Cancel at the wallet sheet	Start payment, then cancel	Driver returns to the ticket, status unchanged	Medium
TC-17	Pay fine	Ticket already paid	Open a ticket marked Paid	Pay button is hidden or disabled	Medium
TC-18	Pay fine	Connection lost during payment	Turn off the network mid-payment, then reconnect	No double charge, status matches the real payment result	High
TC-19	View cars	Owner with several vehicles	Open My Cars	Every registered vehicle is listed with its plate number	High
TC-20	Car info	Open a vehicle	Tap a vehicle in the list	Make, model, year, plate, registration status, linked tickets and accidents shown	High
TC-21	Car info	Search by plate number	Enter a plate that belongs to the driver	Tickets and accident records for that plate are shown	Medium
TC-22	Vehicle registration	Register a new vehicle	Enter plate, VIN, make, model and year, submit	Vehicle appears in My Cars	High
TC-23	Vehicle registration	Duplicate plate number	Register a plate that already exists	Error shown, no duplicate record	Medium
TC-24	My license	View license details	Open License	Number, class, expiry date and status shown	Medium
TC-25	Change location	Update to a valid address	Enter a new address, submit	Address saved, confirmation shown, new tickets use the new location	High

ID	Use case	Test scenario	Steps / input	Expected result	Priority
TC-26	Change location	Empty or invalid address	Submit a blank address, or one with invalid characters	Validation error, old address kept	Medium
TC-27	Issue ticket (admin or cop)	Issue a valid ticket	Log in as cop, enter plate, offence, location and fine, submit	Ticket created and visible in the owner's app immediately	High
TC-28	Issue ticket (admin or cop)	Unknown plate number	Enter a plate that is not registered	Error shown, no ticket created	Medium
TC-29	Issue ticket (admin or cop)	Driver tries the issue screen	Log in as a normal driver, open the issue-ticket URL	Access denied	High
TC-30	Verify ticket	Valid ticket number	Cop enters an existing ticket number	Ticket details and current status shown	High
TC-31	Verify ticket	Ticket number does not exist	Cop enters a made-up ticket number	Not-found message shown	Medium
TC-32	Update ticket status	Change status	Admin changes a ticket to Paid, Cancelled or Disputed	Status saved and the driver sees the new status	High
TC-33	Update ticket status	Driver tries to change status	Log in as a driver, attempt the status update	Action refused, status unchanged	High